Customer Segmentation and Business Insights – Data Science Task #03
Gexton Education – Summer Internship Program

Files
File	Description
`customer_data_raw.csv`	19-row raw dataset (same 19 customers as Task #02, extended with Age/Gender/Income/SpendingScore)
`analysis.py`	Full analysis script: cleaning, EDA, 8+ visualizations, K-Means clustering
`customer_data_cleaned.csv`	Generated after running the script (18 rows, no missing/duplicates)
`customer_data_segmented.csv`	Cleaned data with final cluster assignments
`chart1`-`chart8` `.png`	The 8 required chart types (bar, histogram, pie, box plot x2, scatter x2, heatmap)
`chart9_elbow_method.png`	Elbow method chart used to pick k=3
`chart10_clusters_scatter.png`	Final cluster visualization
`Customer_Segmentation_Report.docx`	Final written report with EDA summary, segmentation results, and business recommendations

Setup
```bash
pip install pandas numpy matplotlib seaborn scikit-learn
```

How to Run
1. Put `customer_data_raw.csv` and `analysis.py` in the same folder.
2. Open the folder in VS Code.
3. Open `analysis.py` and click ▶ Run.
4. Output prints in the terminal; all charts and CSVs are saved to the same folder.

What the Data Contains
19 customers (CustomerID, Age, Gender, AnnualIncome_k, SpendingScore), extended from the same 19 unique customers used in Task #02's sales dataset.

Intentional issues for cleaning practice:
- 1 duplicate row (CUST109)
- 1 missing AnnualIncome value (CUST116)
- 1 genuine income outlier (CUST118 at 150k) — flagged via IQR method, kept (not removed) since it's a real high-value customer, not a data error

What the Script Does
1. **Data Understanding & Cleaning** – shape, dtypes, summary stats, missing

values, duplicates removed, missing income imputed with median, outliers detected via IQR
2. **EDA** – gender distribution, descriptive statistics (mean/median/mode/std/variance), correlation matrix
3. **Visualization** – 8 required chart types (bar, histogram, pie, 2x box plot, 2x scatter, heatmap), each with a one-line insight printed to console
4. **Segmentation (K-Means)** – Elbow Method to pick optimal k, K-Means clustering on Income + Spending Score, cluster profiling and business labeling

Key Results (from this sample)
- 3 customer segments identified:
  - **Cluster 0 – Conservative** (8 customers): moderate-high income, low spending
  - **Cluster 1 – High-Spend/Engaged** (9 customers): younger, high spending score — most valuable segment
  - **Cluster 2 – High Income, Low Engagement** (1 customer, CUST118): VIP-at-risk, highest income but lowest spending

See the Word report for full insights and recommendations.

Note on Sample Size
This dataset (18-19 rows) is intentionally small to mirror Task #02. The full workflow (cleaning → EDA → visualization → clustering → insights) is identical to what you'd run on a larger dataset (e.g. the classic 200-row "Mall Customers" dataset) — only the cluster boundaries would be more statistically reliable with more data.